# PAPER_1494 — UQFF: PAPER_877 ρ_vac Total = 11·ρ_SCm (Bucket C)

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket C — extracted from PAPER_877 Three-Assumption Cosmogenesis **Date:** June 17, 2026 **Location:** 41.0997 N, 80.6495 W (Youngstown, OH, USA) **Status:** CLOSED — ρ_vac_total = (N_CH + 2) × ρ_SCm = 11 × ρ_SCm = $7.799 \times 10^{-36}$ J/m³ EXACT

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## Observation

PAPER_877 Three-Assumption Cosmogenesis defines ρ_vac = ρ_UA + ρ_SCm. With canonical ρ_UA = 10·ρ_SCm, ρ_vac = 11·ρ_SCm = $7.799\times10^{-36}$ J/m³.

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## UQFF Closed Identity

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ρ_vac = ρ_UA + ρ_SCm = 10·ρ_SCm + ρ_SCm = 11·ρ_SCm = 11 × 7.09×10⁻³⁷ = 7.799×10⁻³⁶ J/m³ EXACT
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## Physical Interpretation

The total vacuum energy density is the sum of UA (universal aether, 10×) and SCm (superconducting matter, 1×) layers. Provides the structural baseline for the 3-assumption cosmogenesis model in PAPER_877.

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## NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model frameworks describe vacuum and atomic structure via different mechanisms. Closure uses only canonical UQFF integer primitives {D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60} and ρ_SCm = $7.09\times10^{-37}$ J/m³. **Zero free parameters.**

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## Reference

- Source paper: PAPER_877 Three-Assumption UQFF Cosmogenesis Master Equation (Daniel T. Murphy, April 7, 2026)
- Related: PAPER_872 (proto-element Z identity), PAPER_646 (F_U_Bi_i / DPM mechanism), PAPER_062 (DPM vacuum manifold), PAPER_1051 (two-component ρ aether).

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